UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 7B - Work Measurement: Time Studies II

EIN3241: Human Factors and Ergonomics
Spring 2026

April 1, 2026

Dr. Jad A. Atweh

Admin Issues

- Lab 6 due Friday

What did we learn last time?

- Time Study
- Equipment
- Procedure
- Last step we covered: Calculate elemental times
  - Find the mean elemental times for each element
  - Find the standard deviation for each elemental time

	Element 1	Element 2	Element 3
Cycle 1			
Cycle 2			
Cycle 3			
Cycle 4			
Cycle 5			
Cycle 6			
Average	ET_1	ET_2	ET_3
Standard dev	σ_1	σ_2	σ_3

Time study procedure

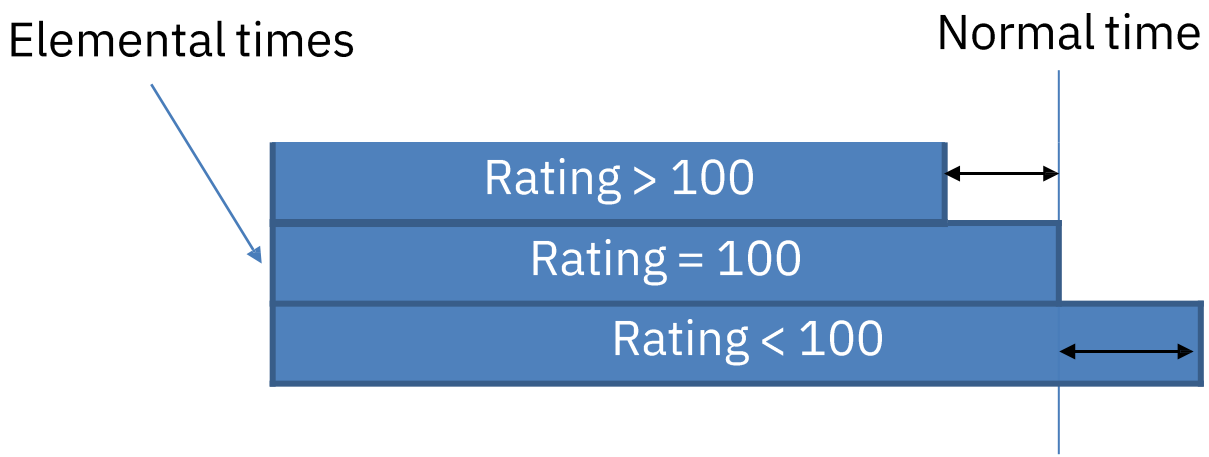
1. Identify breakpoints for your elements
2. Determine number of cycles needed
3. Record the element times
4. Address difficulties in the data and clean up data
5. Calculate the elemental times
6. Performance rate the operator
7. Assign appropriate allowances
8. Calculate the standard time

Time study procedure

1. Identify breakpoints for your elements
2. Determine number of cycles needed
3. Record the element times
4. Address difficulties in the data and clean up data
5. Calculate the elemental times
6. Performance rate the operator
7. Assign appropriate allowances
8. Calculate the standard time

Performance rate the operator

- Even if you tried to find a qualified operator, they may not always be performing the standard method at the standard rate
  - The operator may be going slower
  - The operator may be going faster
- You need to give the operator a performance rating based on how close the operator is to the standard performance
- Different ways to rate performance
  - At each cycle: not practical and hard to do
  - For each element:
    - Typically done for very long and multi-part jobs (cycle time > 30 min)
  - For the whole operation:
    - Typically used for shorter, more homogeneous jobs
- Rating can be open to criticism
  - Different analysts may give different ratings
  - Analyst may be biased



Example 1

- Find the normal time if:
  a. The performance ratings are 95%, 100%, and 115% for E1, E2, and E3, respectively
  b. The overall performance rating is 90%

	Element 1 (s)	Element 2 (s)	Element 3 (s)
Cycle 1	3	6	-
Cycle 2	4	-	-
Cycle 3	3	4	-
Cycle 4	2	6	2
Cycle 5	3	-	-
Cycle 6	4	-	-
Cycle 7	2	6	3
Average	3	5.5	2.5

Example 1a: If you rate each element

	Element 1 (s)	Element 2 (s)	Element 3 (s)
Cycle 1	3	6	-
Cycle 2	4	-	-
Cycle 3	3	4	-
Cycle 4	2	6	2
Cycle 5	3	-	-
Cycle 6	4	-	-
Cycle 7	2	6	3
Average	3	5.5	2.5
Rating	95	100	115

$$Normal\ Element\ Time\ (NT_n) = \overline{ET_n} \times \frac{R_n}{100}$$

$$Normal\ Time\ (NT) = NT_1 + \left(\frac{f_2}{f_1}\right) NT_2 + \cdots + \left(\frac{f_n}{f_1}\right) NT_n$$

f_1 should have the highest frequency

$$NT_1 = 3 \times 0.95 = 2.85\ sec$$
$$NT_2 = 5.5 \times 1 = 5.5\ sec$$
$$NT_3 = 2.5 \times 1.15 = 2.87\ sec$$

$$\rightarrow NT = 2.85 + 5.5\left(\frac{4}{7}\right) + 2.87\left(\frac{2}{7}\right) = 6.78\ sec$$

Example 1b: If you rate the operation as a whole

	Element 1 (s)	Element 2 (s)	Element 3 (s)
Cycle 1	3	6	-
Cycle 2	4	-	-
Cycle 3	3	4	-
Cycle 4	2	6	2
Cycle 5	3	-	-
Cycle 6	4	-	-
Cycle 7	2	6	3
Average	3	5.5	2.5

$$Average\ Elemental\ Time\ (\overline{ET}) = \overline{ET}_1 + \left(\frac{f_2}{f_1}\right)\overline{ET}_2 + \cdots + \left(\frac{f_n}{f_1}\right)\overline{ET}_n$$

$$Normal\ Time\ (NT) = \overline{ET} \times \frac{R}{100}$$

R = 90%

$$\overline{ET} = 3 + 5.5\left(\frac{4}{7}\right) + 2.5\left(\frac{2}{7}\right) = 6.83 \text{ seconds}$$

$$\rightarrow NT = 6.83 \times 0.9 = 6.14 \text{ seconds}$$

How to rate an operator

- Rating methods
  - Speed rating
  - Westinghouse system
- Speed Rating:
  - Consider only the rate of work per unit time
    - Easiest to apply
    - Operator is rated based on the speed of work
    - Analyst needs to have complete knowledge of the job and form a mental model of it
  - Established Benchmarks:

Table 11.1 Speed Rating Guide

Rating	Verbal anchor points	Walking speed (mi/h)	Cards dealt per 1/2 min
0	No activity	0	0
67	Very slow, clumsy	2	35
100	Steady, deliberate	3	52
133	Brisk, businesslike	4	69
167	Very fast, high degree of dexterity	5	87
200	Upper limit for short period	6	104

Westinghouse system

- The Westinghouse system
  - One of the oldest rating systems
  - Considers four factors
    - Skill
    - Effort
    - Conditions
    - Consistency

Westinghouse system

- Skill: proficiency at following a given method
  - Based on operator experience, training, and own aptitude
  - Increases over time
  - Can vary from job to job and over time

Table 11.2 Westinghouse System Skill Ratings

+0.15	A1	Superskill
+0.13	A2	Superskill
+0.11	B1	Excellent
+0.08	B2	Excellent
+0.06	C1	Good
+0.03	C2	Good
0.00	D	Average
−0.05	E1	Fair
−0.10	E2	Fair
−0.16	F1	Poor
−0.22	F2	Poor

Westinghouse system

- Effort: demonstration of the will to work effectively
  - Speed at which the skill is applied

Table 11.3 Westinghouse System Effort Ratings

+0.13	A1	Excessive
+0.12	A2	Excessive
+0.10	B1	Excellent
+0.08	B2	Excellent
+0.05	C1	Good
+0.02	C2	Good
0.00	D	Average
−0.04	E1	Fair
−0.08	E2	Fair
−0.12	F1	Poor
−0.17	F2	Poor

Westinghouse system

- Conditions: include temperature, ventilation, light, and noise
  - Worse than normal conditions result in lower rating

Table 11.4 Westinghouse System Condition Ratings

+0.06	A	Ideal
+0.04	B	Excellent
+0.02	C	Good
0.00	D	Average
−0.03	E	Fair
−0.07	F	Poor

Westinghouse system

- Consistency: how close the element times are to each other in different cycles

Table 11.5 Westinghouse System Consistency Ratings

+0.04	A	Perfect
+0.03	B	Excellent
+0.01	C	Good
0.00	D	Average
−0.02	E	Fair
−0.04	F	Poor

Westinghouse system

- As a final step, the ratings of the four factors are combined to get your total performance rating

Skill	C2	+0.03
Effort	C1	+0.05
Conditions	D	+0.00
Consistency	E	−0.02
Algebraic sum		+0.06
Performance factor		1.06

R = 106

Add one to algebraic sum

Time study procedure

1. Identify breakpoints for your elements
2. Determine number of cycles needed
3. Record the element times
4. Address difficulties in the data and clean up data
5. Calculate the elemental times
6. Performance rate the operator
7. Assign appropriate allowances
8. Calculate the standard time

Execution of study

- **Allowance**: an adjustment made for worker needs and events outside the standard method

Fatigue/rest periods

Unavoidable delays

Personal needs

Time required to learn

Setup and preparation

Abnormal elements

Execution of study

- **Standard time**: time required for a qualified worker to perform a job using a standard method under normal conditions

$$Standard\ time = Normal\ time\ (1 + \frac{Allowance\%}{100})$$

- How to determine how much of an allowance needs to be given?
  - Direct observation: study two or three operators over time
  - Work sampling study (next topic)
  - Experiments and physiological data
- Note that sometimes a process might need to be improved first before giving allowances

Types of allowances

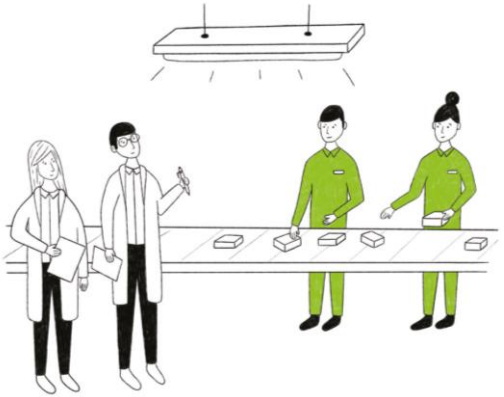
- Constant allowances
  - Constant values added for any type of job
  - Personal needs (5%) + basic fatigue (4%) = 9%
- Variable fatigue allowances
  - Could be physical and/or psychological
  - Main contributors:
    - Abnormal posture
    - Muscular force
    - Atmospheric conditions
    - Noise level
    - Illumination level
    - Visual strain
    - Mental strain
    - Monotony
- Special allowances (Unavoidable delays)

Table 11.9 ILO Recommended Allowances

A. Constant allowances:	
1. Personal allowance	5
2. Basic fatigue allowance	4
B. Variable allowances:	
1. Standing allowance	2
2. Abnormal position allowance:	
a. Slightly awkward	0
b. Awkward (bending)	2
c. Very awkward (lying, stretching)	7
3. Use of force, or muscular energy (lifting, pulling, or pushing):	
Weight lifted, lb:	
5	0
10	1
15	2
20	3
25	4
30	5
35	7
40	9
45	11
50	13
60	17
70	22
4. Bad light:	
a Slightly below recommended	0
b. Well below	2
c. Quite inadequate	5
5. Atmospheric conditions (heat and humidity)—variable	0–100
6. Close attention:	
a. Fairly fine work	0
b. Fine or exacting	2
c. Very fine or very exacting	5
7. Noise level:	
a. Continuous	0
b. Intermittent—loud	2
c. Intermittent—very loud	5
d. High-pitched—loud	5
8. Mental strain:	
a. Fairly complex process	1
b. Complex or wide span of attention	4
c. Very complex	8
9. Monotony:	
a. Low	0
b. Medium	1
c. High	4
10. Tediousness:	
a. Rather tedious	0
b. Tedious	2
c. Very tedious	5

Final notes

- What if you just want the standard time for each element alone?
  - Apply a rating and allowance to each one separately, as applicable
- What do you do if you have more than one worker or multiple stations?
  - Get the standard time for each element
  - Create a precedence chart before calculating the final standard time
- Hawthorne effect: workers change their behavior when they know they are being watched
  - Named after the Hawthorne Works in Chicago where the studies were done
  - Big problem with time studies



UF
Herbert Wertheim
College of Engineering
UNIVERSITY of FLORIDA
LEADING THE CHARGE, CHARGING AHEAD